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Roll No.: 31010924149

Batch - 3

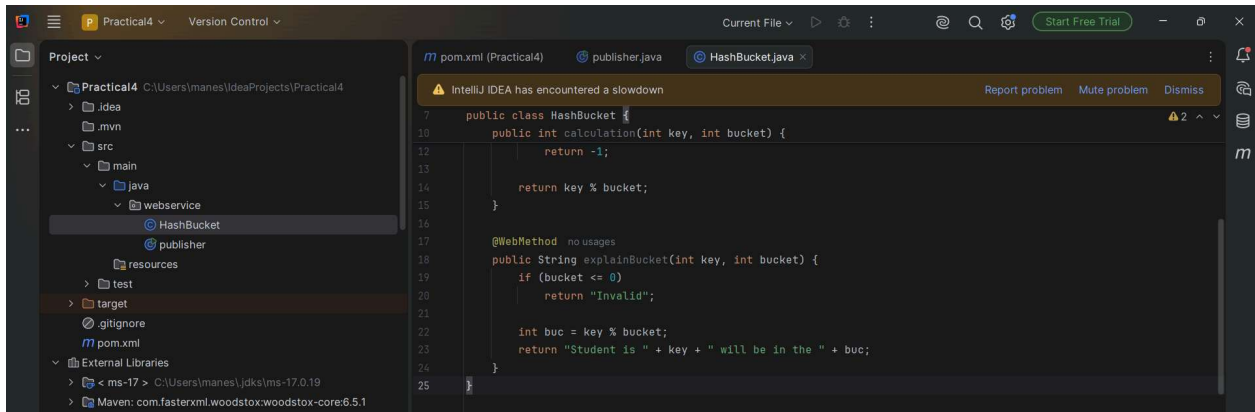
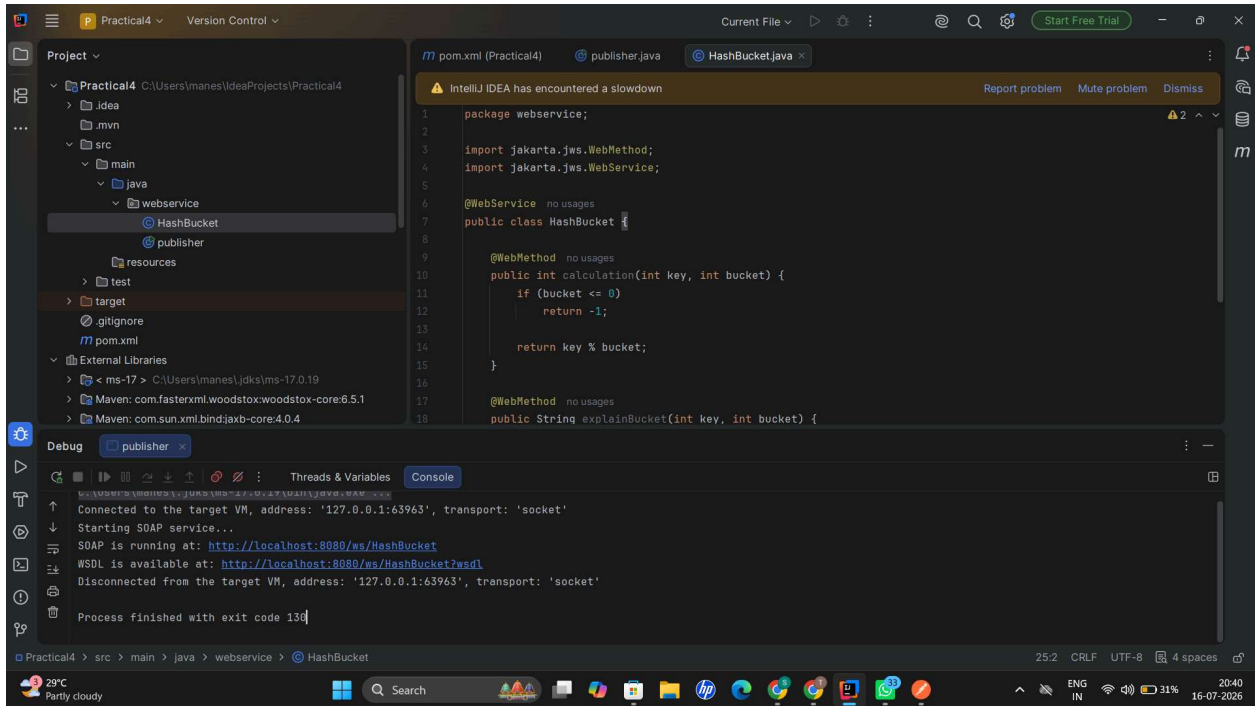
Class: TY Bsc IT - B

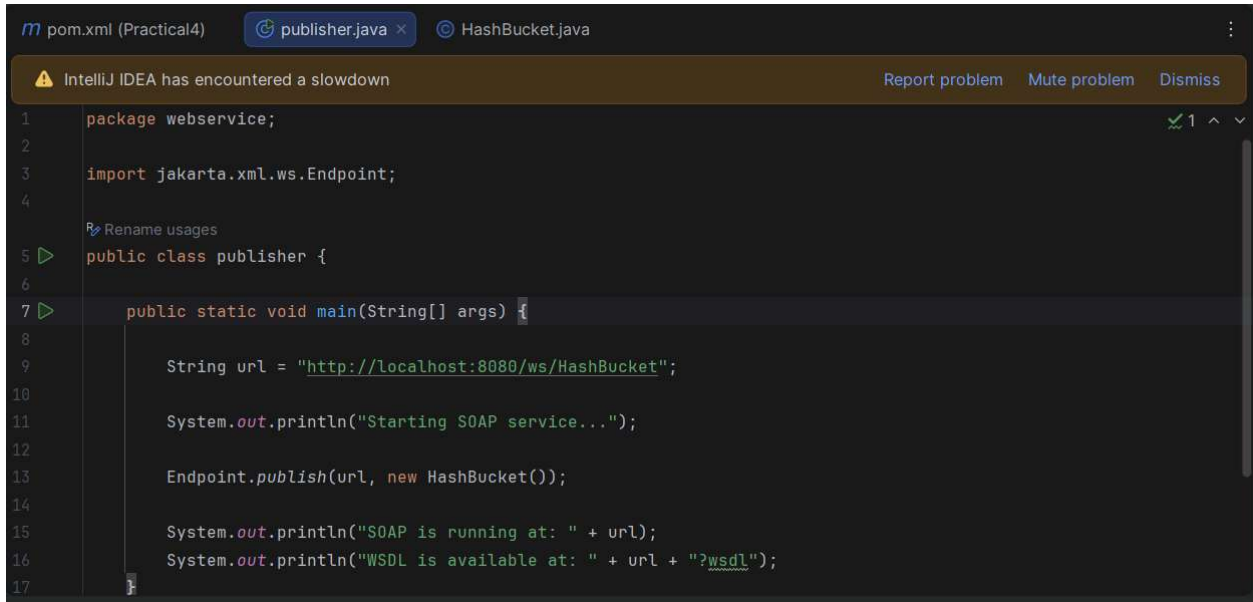
Subject : Web Services

Practical - 4

Steps:

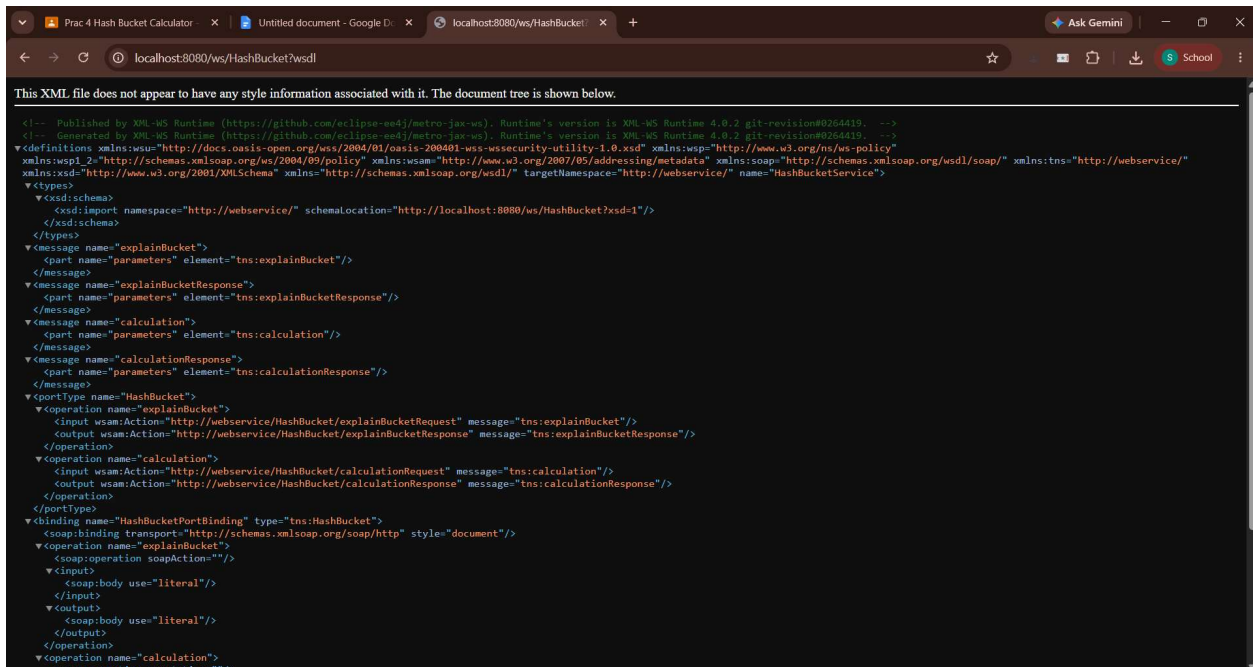
- 1)Open IntelliJ IDEA and create a new **Maven Project**.
- 2)Configure **JDK 17** and update the `pom.xml` with Jakarta SOAP dependencies.
- 3)Reload/Sync the Maven project.
- 4)Create a package named **webservice**.
- 5)Create **HashBucket.java** and implement the SOAP web service methods using `@WebService` and `@WebMethod`.
- 6)Create **publisher.java** and publish the service using `Endpoint.publish()`.
- 7)Run the `publisher.java` class.





```
1 package webservice;
2
3 import jakarta.xml.ws.Endpoint;
4
5 public class publisher {
6
7     public static void main(String[] args) {
8
9         String url = "http://localhost:8080/ws/HashBucket";
10
11         System.out.println("Starting SOAP service...");
12
13         Endpoint.publish(url, new HashBucket());
14
15         System.out.println("SOAP is running at: " + url);
16         System.out.println("WSDL is available at: " + url + "?wsdl");
17     }
18 }
```

8) Verify the WSDL by opening <http://localhost:8080/ws/HashBucket?wsdl> in the browser.



```
<?xml version='1.0' encoding='UTF-8'?>
<!-- Published by XML-WS Runtime (https://github.com/eclipse-ee4j/metro-jax-ws). Runtime's version is XML-WS Runtime 4.0.2 git-revision#0264419. -->
<!-- Generated by XML-WS Runtime (https://github.com/eclipse-ee4j/metro-jax-ws). Runtime's version is XML-WS Runtime 4.0.2 git-revision#0264419. -->
<definitions xmlns:wsu="http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-wssecurity-utility-1.0.xsd" xmlns:wsp="http://www.w3.org/ns/ws-policy"
xmlns:wspl1_2="http://schemas.xmlsoap.org/ws/2004/09/policy" xmlns:wsam="http://www.w3.org/2007/05/addressing/metadata" xmlns:soap="http://schemas.xmlsoap.org/soap/" xmlns:tns="http://webservice/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns="http://schemas.xmlsoap.org/wsdl/" targetNamespace="http://webservice/" name="HashBucketService">
  <types>
    <xsd:schema>
      <xsd:import namespace="http://webservice/" schemaLocation="http://localhost:8080/ws/HashBucket?xsd=1"/>
    </xsd:schema>
  </types>
  <message name="explainBucket">
    <part name="parameters" element="tns:explainBucket"/>
  </message>
  <message name="explainBucketResponse">
    <part name="parameters" element="tns:explainBucketResponse"/>
  </message>
  <message name="calculation">
    <part name="parameters" element="tns:calculation"/>
  </message>
  <message name="calculationResponse">
    <part name="parameters" element="tns:calculationResponse"/>
  </message>
  <portType name="HashBucket">
    <operation name="explainBucket">
      <input wsam:Action="http://webservice/HashBucket/explainBucketRequest" message="tns:explainBucket"/>
      <output wsam:Action="http://webservice/HashBucket/explainBucketResponse" message="tns:explainBucketResponse"/>
    </operation>
    <operation name="calculation">
      <input wsam:Action="http://webservice/HashBucket/calculationRequest" message="tns:calculation"/>
      <output wsam:Action="http://webservice/HashBucket/calculationResponse" message="tns:calculationResponse"/>
    </operation>
  </portType>
  <binding name="HashBucketPortBinding" type="tns:HashBucket">
    <soap:binding transport="http://schemas.xmlsoap.org/soap/http" style="document"/>
    <operation name="explainBucket">
      <soap:operation soapAction="">
        <input>
          <soap:body use="literal"/>
        </input>
        <output>
          <soap:body use="literal"/>
        </output>
      </operation>
    <operation name="calculation">
      <soap:operation soapAction="">
        <input>
          <soap:body use="literal"/>
        </input>
        <output>
          <soap:body use="literal"/>
        </output>
      </operation>
    </binding>
  </definitions>
```

- 9) Open Postman and create a **POST** request.
- 10) Set the URL to <http://localhost:8080/ws/HashBucket>.
- 11) Add the **Content-Type: text/xml** header.
- 12) Send SOAP XML requests to test the `calculation()` and `explainBucket()` methods.
- 13) Verify the responses for both valid and invalid test cases.

sujoy's Workspace

Search

Overview POST http://localhost:8080/ HashBucket

POST http://localhost:8080/ws/HashBucket

Docs Params Authorization Headers 9 Body Scripts Tests Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <soapenv:Envelope
3   xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
4   xmlns:tns="http://webservice/"
5
6   <soapenv:Header/>
7
8   <soapenv:Body>
9
10    <tns:calculation>
11
12      <arg0>13</arg0>
13
14      <arg1>5</arg1>
15
16    </tns:calculation>
```

Body Cookies Headers 3 Test Results

XML Preview Visualize

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
3   <S:Body>
4     <ns2:calculationResponse xmlns:ns2="http://webservice/"
5       <return>3</return>
6     </ns2:calculationResponse>
7   </S:Body>
8 </S:Envelope>
```

sujoy's Workspace

Search

Overview POST http://localhost:8080/ HashBucket

POST http://localhost:8080/ws/HashBucket

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none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <soapenv:Envelope
3   xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
4   xmlns:tns="http://webservice/"
5
6   <soapenv:Header/>
7
8   <soapenv:Body>
9
10    <tns:calculation>
11
12      <arg0>5</arg0>
13
14      <arg1>0</arg1>
15
16    </tns:calculation>
```

Body Cookies Headers 3 Test Results

XML Preview Visualize

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/"
3   <S:Body>
4     <ns2:calculationResponse xmlns:ns2="http://webservice/"
5       <return>-1</return>
6     </ns2:calculationResponse>
7   </S:Body>
8 </S:Envelope>
```

sujoy's Workspace

+

+

+

+

COLLECTIONS

My Collection

Overview

POST http://localhost:8080/

+

http://localhost:8080/ws/HashBucket

POST

http://localhost:8080/ws/HashBucket

Docs

Params

Authorization

Headers 9

Body

Scripts

Tests

Settings

☐ none

☐ form-data

☐ x-www-form-urlencoded

☒ raw

☐ binary

☐ GraphQL

☐ JSON

⚠

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

<soapenv:Header/>

<soapenv:Body>

<tns:explainBucket>

<arg0>13</arg0>

<arg1>5</arg1>

</tns:explainBucket>

</soapenv:Body>

</soapenv:Envelope>

Body

Cookies

Headers 3

Test Results

🔄

XML

Preview

Visualize

⌵

1

2

3

4

5

6

7

8

<?xml version='1.0' encoding='UTF-8'?>

<S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/">

<S:Body>

<ns2:explainBucketResponse xmlns:ns2="http://webservice/">

<return>Student is 13 will be in the 3</return>

</ns2:explainBucketResponse>

</S:Body>

</S:Envelope>

sujoy's Workspace

COLLECTIONS

- My Collection

Overview POST http://localhost:8080/ HashBucket

POST http://localhost:8080/ws/HashBucket

Docs Params Authorization Headers 9 Body Scripts Tests Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
5
6     <soapenv:Header/>
7
8     <soapenv:Body>
9
10        <tns:explainBucket>
11            <arg0>5</arg0>
12            <arg1>0</arg1>
13        </tns:explainBucket>
14    </soapenv:Body>
15
16</soapenv:Envelope>
```

Body Cookies Headers 3 Test Results

XML Preview Visualize

```
1 <?xml version='1.0' encoding='UTF-8'?>
2 <S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/">
3     <S:Body>
4         <ns2:explainBucketResponse xmlns:ns2="http://webservice/">
5             <return>Invalid</return>
6         </ns2:explainBucketResponse>
7     </S:Body>
8 </S:Envelope>
```